

Benchmarking Factual Accuracy of Large Language Models Across Scientific Domains

A Review of Methods, Benchmarks, and Open Problems

Abstract

Large language models (LLMs) are increasingly consulted for scientific information spanning biomedicine, chemistry, physics, and the life sciences, yet their propensity to generate fluent but unsubstantiated statements — commonly termed “hallucination” — raises concerns about their reliability as scientific knowledge sources (Ji et al., 2023). This paper synthesizes the current landscape of benchmarks and methodologies used to evaluate the factual accuracy of LLMs across scientific domains. We review general-purpose factuality benchmarks such as TruthfulQA (Lin et al., 2022) and SimpleQA (Wei et al., 2024), domain-specific evaluations including PubMedQA (Jin et al., 2019) and MultiMedQA (Singhal et al., 2023), and graduate-level scientific reasoning benchmarks such as GPQA (Rein et al., 2023). We further examine claim-verification frameworks grounded in the scientific literature, notably SciFact (Wadden et al., 2020), and mitigation strategies including retrieval-augmented generation (Lewis et al., 2020). Drawing on this literature, we characterize recurring findings: LLM factual performance varies substantially across scientific sub-domains, correlates with model scale and fine-tuning strategy, and remains markedly below expert human performance on adversarially constructed, “Google-proof” questions (Rein et al., 2023). We then identify structural limitations of existing benchmarks, including data contamination, benchmark saturation, static ground truth in rapidly evolving fields, and the high cost of expert annotation. We conclude by outlining research gaps — particularly the need for dynamic, contamination-resistant, multimodal, and uncertainty-aware evaluation frameworks — and propose directions for future work aimed at making LLM factuality assessment in scientific domains more rigorous, reproducible, and clinically and scientifically actionable.

Keywords: large language models; factual accuracy; hallucination; scientific benchmarking; biomedical question answering; retrieval-augmented generation; fact verification; natural language processing

1. Introduction

The rapid diffusion of large language models such as GPT-4, PaLM, and their successors into scientific and clinical workflows has been accompanied by growing scrutiny of whether these systems can be trusted to state facts correctly (Singhal et al., 2023). Because LLMs are trained to predict plausible continuations of text rather than to verify propositions against a ground truth, they are structurally prone to producing hallucinations: content that is fluent and internally coherent but factually incorrect or unsupported by evidence (Ji et al., 2023). In general-purpose settings, such errors may be merely inconvenient; in scientific and biomedical settings, where practitioners, researchers, and patients may rely on model output to inform decisions, factual errors carry a materially higher cost.

Benchmarking factual accuracy is therefore a foundational research problem. Unlike many natural language processing tasks in which performance can be assessed with fluency- or coherence-based metrics,

factuality evaluation requires an explicit reference to ground-truth knowledge, an operational definition of what counts as an error, and — in scientific domains particularly — access to domain expertise capable of adjudicating claims that are not settled by simple lookup (Wadden et al., 2020). Over the past several years, the research community has produced an increasingly diverse set of benchmarks intended to interrogate this capability, ranging from short-form, adversarially collected question sets such as TruthfulQA (Lin et al., 2022) and SimpleQA (Wei et al., 2024), to domain-specific suites in biomedicine such as PubMedQA (Jin et al., 2019) and MultiMedQA (Singhal et al., 2023), to graduate-level, expert-authored question sets spanning physics, chemistry, and biology such as GPQA (Rein et al., 2023).

This paper provides a synthesis of this literature with a specific focus on scientific domains. We ask three organizing questions: (1) how has the field operationalized “factual accuracy” for LLMs across different scientific sub-fields; (2) what do existing empirical results reveal about the strengths and weaknesses of current models; and (3) what methodological and structural gaps remain in the benchmarking ecosystem. The remainder of the paper proceeds as follows. Section 2 reviews background literature on hallucination and general factuality evaluation. Section 3 discusses domain-specific benchmarking efforts in biomedicine, the physical sciences, and scientific claim verification. Section 4 surveys current approaches to improving and measuring factuality, including retrieval augmentation and human evaluation frameworks. Section 5 discusses research challenges and limitations, and Section 6 identifies research gaps and future directions. Section 7 concludes.

2. Background and Related Work

2.1 Hallucination as a Conceptual Foundation

The concept of hallucination in natural language generation predates the current generation of LLMs and was originally developed to describe unfaithful or unsupported content produced by sequence-to-sequence models in tasks such as summarization and data-to-text generation (Ji et al., 2023). Ji et al. (2023) distinguish between intrinsic hallucinations, in which generated content directly contradicts the source material, and extrinsic hallucinations, in which generated content cannot be verified against the source at all. This taxonomy has proven durable as a framework for factuality research in LLMs, where the “source” may be an explicit retrieved document, an implicit body of world knowledge encoded in model parameters, or a domain-specific corpus such as the peer-reviewed scientific literature. The survey further catalogues evaluation methods spanning statistical metrics, model-based metrics, and human evaluation, noting that no single metric class fully captures the phenomenon, a conclusion that motivates much of the benchmark proliferation discussed below.

2.2 General-Purpose Factuality Benchmarks

TruthfulQA (Lin et al., 2022) was among the first large-scale benchmarks designed explicitly to probe factuality rather than fluency. The benchmark comprises questions crafted so that some humans would answer falsely due to common misconceptions, and the authors found that larger models were, at the time, not necessarily more truthful, since scale can amplify the reproduction of popular but false beliefs learned

from training text. This finding motivated a shift away from the assumption that scaling model size alone would resolve factuality problems, and toward fine-tuning objectives and evaluation protocols explicitly targeting truthfulness.

More recently, OpenAI’s SimpleQA (Wei et al., 2024) was introduced as a short-form factuality benchmark comprising several thousand adversarially collected, fact-seeking questions with single, unambiguous answers, designed in part to replace earlier benchmarks such as TriviaQA and Natural Questions that had become saturated by the performance of frontier models. A defining feature of SimpleQA is its grading scheme, which distinguishes between incorrect answers and appropriate abstentions, operationalizing the idea that a well-calibrated model should decline to answer questions it is not confident about rather than confabulate a response. Complementary efforts such as FELM (Chen et al., 2023) extend fine-grained factuality annotation beyond world knowledge into domains including mathematics and reasoning, annotating factuality at the level of text segments rather than whole responses, which allows finer localization of errors and supports development of automated factuality evaluators. Collectively, these general-purpose benchmarks establish the methodological vocabulary — short-form versus long-form factuality, abstention-aware grading, and segment-level annotation — that domain-specific scientific benchmarks subsequently adapt.

2.3 Toward Domain-Specific Evaluation

General-purpose factuality benchmarks are necessary but insufficient for evaluating scientific reliability, because scientific domains impose distinctive requirements: claims must often be adjudicated against a technical literature rather than common-sense knowledge, correctness may be probabilistic or context-dependent (as in biomedical research findings), and the cost of an undetected error can be substantially higher. This has motivated a parallel research thread constructing domain-specific benchmarks in biomedicine, chemistry, physics, and biology, which we discuss in detail in Section 3.

3. Factual Accuracy Across Scientific Domains

3.1 Biomedical and Clinical Question Answering

Biomedicine has become the most extensively benchmarked scientific domain for LLM factuality, in part because of the direct relevance of clinical decision support to patient safety. PubMedQA (Jin et al., 2019) established an early paradigm in which models must answer yes/no/maybe research questions using the corresponding PubMed abstract as context, requiring genuine reasoning over quantitative biomedical findings rather than superficial pattern matching; on this task, the original benchmark reported single-human performance of 78.0% accuracy against a majority-class baseline of 55.2%, illustrating that even expert-level performance on this task is far from perfect, underscoring the intrinsic difficulty of biomedical claim adjudication.

Building on datasets such as PubMedQA, MedQA, and MedMCQA, Singhal et al. (2023) introduced MultiMedQA, a benchmark combining six existing medical question-answering datasets with a new consumer-facing dataset, HealthSearchQA, and proposed a human evaluation framework assessing model

answers along axes including factuality, comprehension, reasoning, potential harm, and bias. Using this framework, the authors evaluated PaLM and its instruction-tuned variant Flan-PaLM, and introduced instruction prompt tuning to produce Med-PaLM. Flan-PaLM achieved state-of-the-art accuracy on every MultiMedQA multiple-choice dataset, including 67.6% accuracy on MedQA (USMLE-style questions), a gain of more than 17 percentage points over the prior state of the art (Singhal et al., 2023). Despite this quantitative gain, human evaluation revealed that Flan-PaLM’s long-form answers to consumer medical questions contained meaningful factual and reasoning gaps relative to clinician-written answers, demonstrating that strong performance on multiple-choice factual recall does not automatically translate into safe, factually grounded long-form generation.

A follow-on system, Med-PaLM 2, combined an improved base model with medical domain fine-tuning and new prompting strategies, including ensemble refinement and chain-of-retrieval reasoning, and improved upon Med-PaLM by more than 19 percentage points on MedQA while approaching or exceeding prior state-of-the-art results on MedMCQA, PubMedQA, and MMLU clinical topics (Singhal et al., 2025). Notably, the authors caution that multiple-choice benchmark performance, while a useful proxy for encoded medical knowledge, does not capture a model’s ability to generate safe, factually grounded answers to the nuanced, open-ended questions characteristic of real clinical use, reinforcing a theme recurrent across domains: benchmark accuracy and deployment-relevant factual reliability are related but distinct constructs.

3.2 Graduate-Level Reasoning in the Physical and Life Sciences

Outside biomedicine, GPQA (Rein et al., 2023) provides one of the most rigorous tests of scientific factual and reasoning capability. The benchmark comprises 448 multiple-choice questions in biology, physics, and chemistry, written and validated by domain experts holding or pursuing doctorates in the corresponding fields, and is explicitly designed to be “Google-proof”: highly skilled non-expert validators with unrestricted internet access and an average of over 30 minutes per question achieved only 34% accuracy, while domain experts achieved 65% accuracy (74% after conservatively discounting flawed items) (Rein et al., 2023). At the time of the benchmark’s release, the strongest GPT-4-based baseline achieved only 39% accuracy in the closed-book setting, well below expert performance, though the authors and subsequent leaderboards report that frontier models have continued to close this gap rapidly in later model generations. The explicit design goal of GPQA — producing questions that remain difficult even for skilled humans with search access — makes it a particularly informative instrument for tracking whether improvements in LLM benchmark scores reflect genuine scientific reasoning or exploitation of superficial cues, and its rapid approach toward saturation on some leaderboards has itself become a data point in discussions of benchmark validity (see Section 5).

3.3 Verifying Claims Against the Scientific Literature

A distinct but complementary strand of work treats factuality not as a question-answering problem but as a claim-verification problem: given an assertion, does the peer-reviewed literature support or refute it? Wadden et al. (2020) introduced SciFact, a dataset of 1,409 expert-written scientific claims paired with a corpus of 5,183 abstracts, each claim requiring a system to identify relevant abstracts, classify them as

supporting or refuting the claim, and provide rationale sentences justifying the decision. Because the claims are grounded in actual biomedical literature and require sentence-level evidentiary rationale rather than a single label, SciFact operationalizes a stricter and more literature-grounded notion of factuality than multiple-choice question answering, and has subsequently been extended to larger, open-domain retrieval settings via SciFact-Open, which situates claim verification against a corpus of roughly 500,000 documents rather than a curated abstract set. This claim-verification paradigm is directly relevant to LLM evaluation because it mirrors the structure of retrieval-augmented factual generation: a system must not only produce a plausible-sounding statement but ground that statement in identifiable, checkable literature evidence.

4. Current Approaches and Developments

4.1 Retrieval-Augmented Generation

The dominant technical strategy for improving — and, relatedly, for evaluating — factual accuracy is retrieval-augmented generation (RAG), introduced by Lewis et al. (2020), which combines a parametric sequence-to-sequence model with a differentiable dense retriever over a non-parametric knowledge index. The central motivation is that purely parametric models store factual knowledge implicitly and imprecisely in their weights, limiting their ability to access, update, or provide provenance for specific facts, whereas coupling generation to an explicit retrieval step allows a model to condition its output on retrieved passages and, in principle, cite the evidence underlying a claim (Lewis et al., 2020). In scientific domains, RAG pipelines built over corpora such as PubMed abstracts have become a standard architecture for grounding LLM outputs in biomedical literature (as reflected in datasets such as PubMedQA and BioASQ being used within retrieval-augmented evaluation frameworks), and the approach is widely regarded as a partial but incomplete mitigation for hallucination, since retrieval quality, passage relevance, and the model’s faithfulness to retrieved evidence each introduce their own failure modes.

4.2 Fine-Grained and Segment-Level Factuality Evaluation

A second methodological development is the move from whole-response factuality judgments toward fine-grained, segment-level annotation. FELM (Chen et al., 2023) annotates factuality at the level of text segments within long-form LLM responses, supplementing binary labels with predefined error-type taxonomies and reference links that support or contradict each segment, and uses this resource to benchmark LLM-based factuality evaluators themselves, including variants augmented with retrieval and chain-of-thought prompting. The authors report that retrieval augmentation improves the reliability of automated factuality evaluators relative to vanilla LLM judges, but that current LLM-based evaluators still fall short of the reliability required for fully automated fact-checking, motivating continued reliance on human-in-the-loop evaluation for high-stakes scientific and biomedical claims.

4.3 Structured Human Evaluation Frameworks

Because automated metrics remain imperfect proxies for expert judgment, several biomedical benchmarking efforts have paired automated multiple-choice evaluation with structured human evaluation. Singhal et al. (2023) propose a multi-axis clinician evaluation protocol assessing factuality, comprehension,

retrieval of relevant knowledge, reasoning, likelihood of harm, and bias, applied by physicians to long-form model answers. This dual-track approach — automated benchmarks for scalable, reproducible comparison across models, and structured expert evaluation for ecologically valid assessment of deployment-relevant answers — has become something of a de facto standard for high-stakes scientific evaluation and is echoed in Med-PaLM 2’s continued reliance on physician-panel review even after substantial gains on multiple-choice benchmarks (Singhal et al., 2025).

5. Research Challenges and Limitations

Despite substantial progress, several structural challenges limit the interpretability and durability of current factuality benchmarks in scientific domains.

Benchmark saturation and contamination. As frontier models are trained on ever-larger web-scale corpora, there is a persistent risk that benchmark questions or closely paraphrased variants appear in training data, inflating apparent performance without corresponding gains in genuine factual competence. GPQA was explicitly designed to resist this failure mode by requiring “Google-proof” difficulty (Rein et al., 2023), yet leaderboard tracking has shown some benchmark subsets approaching saturation as newer model generations are released, illustrating that even carefully constructed benchmarks have a limited useful lifespan.

Static ground truth in a dynamic evidence base. Scientific knowledge, and biomedical knowledge in particular, evolves as new studies are published and prior findings are revised or overturned. Benchmarks constructed from a fixed literature snapshot, such as SciFact’s corpus of abstracts, can become stale, and a claim correctly labeled as “supported” at benchmark construction time may later be contradicted by subsequent research (Wadden et al., 2020). This tension between the need for stable, reusable evaluation sets and the fluidity of scientific consensus is not yet fully resolved by any existing benchmark design.

Cost and scalability of expert annotation. High-quality scientific benchmarks depend on domain-expert authorship and adjudication — GPQA’s questions required PhD-level domain experts for both construction and validation (Rein et al., 2023), and Med-PaLM’s evaluation protocol depends on physician panels for long-form answer review (Singhal et al., 2023). This makes such benchmarks expensive and slow to construct or refresh relative to the pace of model development, and may bias coverage toward sub-domains where expert annotators are more readily available.

Gap between multiple-choice accuracy and long-form factual reliability. A recurring finding across the reviewed literature is that strong performance on multiple-choice factual recall tasks does not guarantee factually reliable long-form generation. Singhal et al. (2023) found meaningful gaps between Flan-PaLM’s strong MultiMedQA multiple-choice performance and clinician assessments of its open-ended answers, and this discrepancy is echoed in the broader hallucination literature’s distinction between different generation settings (Ji et al., 2023). Benchmarks that rely predominantly on multiple-choice formats may therefore systematically overstate real-world factual reliability.

Imperfect automated factuality evaluators. Because human evaluation does not scale to the volume of model outputs requiring assessment, there is strong incentive to automate factuality judgment using LLM-based evaluators. However, Chen et al. (2023) show that even retrieval-augmented LLM evaluators fall short of the reliability needed for unsupervised fact-checking, meaning that automated benchmarking pipelines themselves require ongoing validation against human judgment, adding a layer of meta-evaluation complexity to the field.

6. Research Gaps and Future Directions

Several directions appear promising for advancing the rigor and practical relevance of factuality benchmarking in scientific domains.

Dynamic and continuously updated benchmarks. Following the precedent of dynamic question-answering platforms that refresh questions on a rolling basis, scientific factuality benchmarks could be periodically regenerated from newly published literature to reduce staleness and contamination risk, particularly in fast-moving biomedical sub-fields.

Calibration- and abstention-aware evaluation. Building on SimpleQA’s explicit treatment of abstention as distinct from error (Wei et al., 2024), future scientific benchmarks would benefit from evaluation protocols that reward well-calibrated uncertainty expression, since a model that declines to answer beyond its evidentiary basis may be more useful and safer in scientific and clinical contexts than one that always produces a confident but occasionally fabricated answer.

Finer-grained, evidence-linked annotation across domains. The segment-level, evidence-linked annotation approach pioneered by FELM (Chen et al., 2023) and the rationale-based structure of SciFact (Wadden et al., 2020) could be extended more systematically into physics, chemistry, and engineering domains, which currently rely more heavily on multiple-choice formats such as GPQA and consequently offer less insight into the localization and provenance of factual errors in long-form scientific explanation.

Integrated benchmarking of retrieval and generation. Because retrieval-augmented generation is now the dominant mitigation strategy (Lewis et al., 2020), factuality benchmarks would benefit from evaluation protocols that jointly assess retrieval precision, evidence faithfulness, and final-answer correctness, rather than treating end-to-end accuracy as a single opaque metric, so that failures can be attributed to retrieval versus generation components.

Cross-domain and multimodal extension. Much of the most rigorous domain-specific benchmarking work to date is concentrated in biomedicine, with physics, chemistry, and biology comparatively well served only by GPQA’s multiple-choice format. Extending literature-grounded claim verification and fine-grained factuality annotation to these domains, and to multimodal scientific content such as figures, chemical structures, and experimental images, represents a substantial open area for future benchmark development.

Standardized meta-evaluation of automated judges. Given that LLM-based factuality evaluators are themselves imperfect (Chen et al., 2023), the field would benefit from standardized protocols for validating

automated evaluators against expert human judgment across scientific domains, analogous to the human evaluation frameworks already established in biomedicine (Singhal et al., 2023), so that claims of improved factuality are not circularly validated by similarly unreliable automated judges.

7. Conclusion

Benchmarking the factual accuracy of large language models across scientific domains has matured considerably over the past several years, evolving from general-purpose truthfulness probes such as TruthfulQA (Lin et al., 2022) toward increasingly rigorous, domain-specific instruments including PubMedQA (Jin et al., 2019), MultiMedQA (Singhal et al., 2023), GPQA (Rein et al., 2023), and SciFact (Wadden et al., 2020). Across this literature, a consistent picture emerges: model factual performance has improved substantially with scale, instruction tuning, and domain-specific fine-tuning, yet remains below expert human performance on the most rigorously constructed benchmarks, and multiple-choice benchmark accuracy is only an imperfect proxy for the factual reliability of long-form scientific and clinical generation. Mitigation strategies such as retrieval-augmented generation (Lewis et al., 2020) and structured human evaluation frameworks (Singhal et al., 2023) have made meaningful progress but do not fully resolve the underlying tension between the statistical, next-token-prediction foundations of LLMs and the strict evidentiary demands of scientific claims (Ji et al., 2023). Continued progress will likely depend on dynamic, contamination-resistant, evidence-linked, and calibration-aware benchmarks that treat factuality not as a single scalar score but as a multidimensional property spanning retrieval, reasoning, evidence attribution, and appropriate epistemic humility. As LLMs become further embedded in scientific research and clinical practice, the rigor of the benchmarks used to evaluate them will remain a direct determinant of how safely and productively these systems can be deployed.

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